

THE SCOTS COLLEGE
Sydney

2006
HSC TRIAL
EXAMINATION

Extension One Mathematics

General Instructions

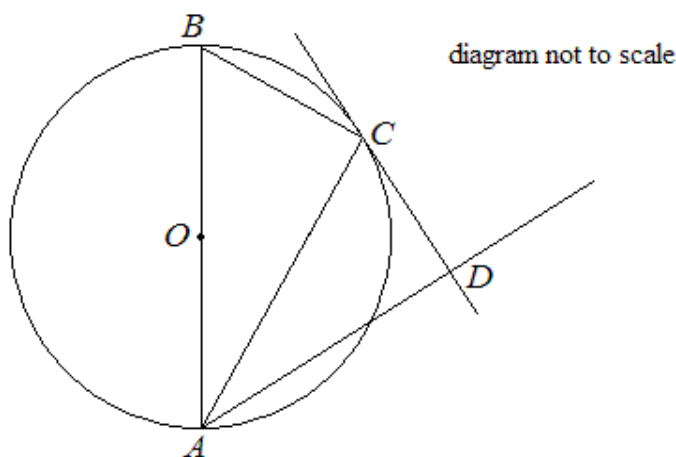
- Working time – 120 Minutes
+ 5 minutes reading time
 - Write using black or blue pen
 - Board-approved calculators may be used
 - Start a new booklet for each question
 - A table of Standard Integrals is attached
 - All necessary working should be shown in every question
- **Total marks - 84**
 - Attempt all questions
 - Start a new booklet for every question

QUESTION ONE**[12 MARKS]****START A NEW BOOKLET**

- a. Solve the inequality $\frac{x^2 - 4}{x} \geq 3$. 3
- b. Let A(-3,6) and B(1,10) be the points on the number plane. Find the coordinates of the point C which divides the interval AB externally in the ratio 5:3. 2
- c. If α, β and γ are the roots of $3x^3 + 5x^2 - 7x + 4 = 0$ find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. 2
- d. Use the substitution $u = 1 - x^3$ to evaluate $\int_0^1 x^2(1 - x^3)^4 dx$. 2
- e. Find the acute angle between the lines $x + 3y - 4 = 0$ and $2x - 5y = 0$. 3
Answer to the nearest degree.

QUESTION TWO**[12 MARKS]****START A NEW BOOKLET**

- a. Find the exact value of $\int_0^{\frac{\pi}{4}} \cos^2 \frac{x}{2} dx$. 3
- b. Find $\int_0^{\frac{1}{5}} \frac{2dx}{\sqrt{16 - 100x^2}}$ 3
- c. Consider the function $y = \frac{\pi}{2} + \cos^{-1} \frac{x}{2}$.
(i) State the domain of this function. 1
(ii) State the range of this function. 1
(iii) Sketch the graph of this function. 1
- d.



In the diagram AOB is the diameter of a circle centre O, and C is the point of contact of the tangent DC such that AC bisects $\angle DAB$.
Prove that AD is perpendicular to DC.

QUESTION THREE**[12 MARKS]****START A NEW BOOKLET**

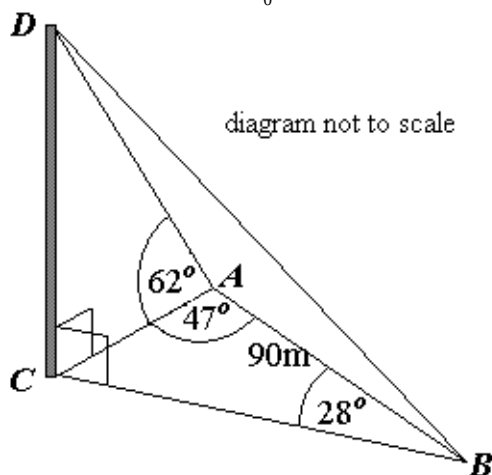
- a. (i) Using the identities for the expansion of $\sin(A+B)$, $\sin 2A$ and $\cos 2A$, prove that $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$. 2

- (ii) Hence solve the equation $3\sin\theta - 4\sin^3\theta = -1$ for $0 \leq \theta \leq 2\pi$. 2

- b. (i) Find $\frac{d}{dx}(x \tan^{-1} x)$. 1

- (ii) Hence find the exact value of $\int_0^1 \tan^{-1} x dx$. 3

c.



A person sees the top of a pole from 2 points A and B both on flat ground. The angle of elevation from point A to the top of the pole D is 62° .

If $\angle ABC = 28^\circ$, $\angle BAC = 47^\circ$ and A and B are 90 metres apart, find;

- i) how far A is from the base of the pole to the nearest metre. 2

- ii) the height of the pole, correct to 3 significant figures. 2

QUESTION FOUR**[12 MARKS]****START A NEW BOOKLET**

- a. Rewrite $\sqrt{3} \sin x - \cos x$ in the form $R \sin(x - \alpha)$. Hence solve $\sqrt{3} \sin x - \cos x = 1$ for $0 \leq x \leq 2\pi$ 3

- b. Find the constant term of the expansion of $\left(2x - \frac{1}{x^3}\right)^{20}$ 2

- c. (i) Show that the normal to the parabola $x^2 = 4ay$ at the point $(2at, at^2)$ has the equation $x + ty = 2at + at^3$ 2

- (ii) Hence show that there is only one normal which passes through the focus. 2

- d. It is known that the equation $e^{-x^2} - 5x^2 - 0.99 = 0$ has a positive root close to the origin. Attempt to find the root using one application of Newton's method, starting with $x = 0$ as the first approximation and hence explain why Newton's method fails with $x = 0$ as the first approximation. 3

QUESTION FIVE**[12 MARKS]****START A NEW BOOKLET**

- a. Prove by Mathematical Induction that $5^{2n} - 1$ is divisible by 6 when n is a positive integer. 4
- b. A spherical balloon is being inflated at a constant rate of $1000\text{cm}^3\text{s}^{-1}$.
Given that a sphere has volume $V = \frac{4}{3}\pi r^3$ and surface area $A = 4\pi r^2$ find:
- (i) An expression for the instantaneous rate of change of the radius in terms of r . (find $\frac{dr}{dt}$) 3
- (ii) The rate of change of the surface area of the balloon when the radius is 10cm. 2
- c. If ${}^nC_r = {}^nC_{r+1}$, where n and r are positive integers, show that n is odd. 3

QUESTION SIX**[12 MARKS]****START A NEW BOOKLET**

- a. A particle's motion is defined by the equation $v^2 = 12 + 4x - x^2$ where x is the displacement in metres of the particle from the origin and v is the velocity in ms^{-1} . Initially the particle is 6 metres to the right of the origin.
- (i) Show that the particle is moving in Simple Harmonic Motion. 1
- (ii) Find the centre, the period and the amplitude of the motion. 3
- (iii) The displacement of the particle at any time t is given by the equation $x = a \sin(nt + \theta) + b$. Find the values of θ and b , given $0 \leq \theta \leq 2\pi$. 2
- b. Newton's Law of Cooling states that the rate of change in the temperature, T of a body is proportional to the difference between the temperature of the body and the surrounding temperature, P .
- (i) If A and k are constants, show that the equation $T = P + Ae^{kt}$ satisfies Newton's Law of Cooling. 2
- (ii) On a day when the air temperature was 18°C , water in a container is heated to 100°C . Three minutes after being taken off the heat the temperature has dropped to 87°C , calculate A and k (to 4 d.p). 2
- (iii) If the temperature has to be below 70°C to wash your hands in the container how long, to the nearest minute, will you have to wait? 2

- a.** A projectile is fired horizontally with speed $v \text{ ms}^{-1}$ from a point $h \text{ m}$ above the horizontal ground. Let acceleration due to gravity be g

(i) Prove that it will reach the ground after $\sqrt{\frac{2h}{g}}$ seconds. 2

(ii) If it does so at an angle of 60° to the horizontal, prove that $3v^2 = 2gh$. 3

Hint: $\frac{dy}{dx} = \tan 120^\circ$

- b.** Given $y = \log_e(e^x \sin^2 x)$

(i) Show that $\frac{dy}{dx} = 1 + 2 \cot x$. 1

(ii) Prove that the equation of the normal at $x = \frac{\pi}{2}$ is given by $x + y = \pi$. 3

c. Find $\int \frac{x dx}{(25 + x^2)^{\frac{3}{2}}}$ using the substitution $x = 5 \tan \theta$ 3

Question one

a) $\frac{x^2-4}{x} \geq 3$

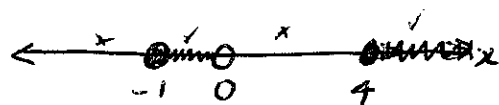
$$x(x^2-4) \geq 3x^2$$

$$x^3-4x-3x^2 \geq 0$$

$$x(x^2-3x-4) \geq 0$$

$$x(x-4)(x+1) \geq 0$$

$$x \neq 0$$



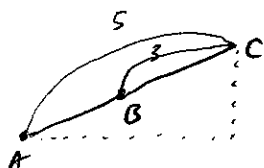
$$-1 < x < 0 \text{ and } x \geq 4$$

b) $A(x_1, y_1) = A(-3, 6)$ $B(x_2, y_2) = B(1, 10)$ $m = -5, n = 3$

$$C = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$= \left(\frac{-5 \cdot 1 + 3 \cdot (-3)}{-5+3}, \frac{-5 \cdot 10 + 3 \cdot 6}{-5+3} \right)$$

$$= (7, 16)$$



c) $\frac{a}{3x^3} + \frac{b}{5x^2} - \frac{c}{7x} + \frac{d}{4} = 0$

now $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\delta} = \frac{\alpha\beta + \alpha\delta + \beta\delta}{\alpha\beta\delta}$

$$= \frac{c/a}{-d/a}$$

$$= -\frac{c}{d}$$

$$= -\frac{7}{4}$$

d) $\int_0^1 x^2(1-x^3)^4 dx$

let $u = 1-x^3$

$$\frac{du}{dx} = -3x^2$$

$$du = -3x^2 dx$$

$$-\frac{1}{3} du = x^2 dx$$

when $x=0, x=1$
 $u=1, u=0$

$$\int_0^1 x^2(1-x^3)^4 dx = -\frac{1}{3} \int_1^0 u^4 du$$

$$= -\frac{1}{3} \left[\frac{u^5}{5} \right]_1^0$$

$$= -\frac{1}{3} \left(0 - \frac{1}{5} \right)$$

$$= \frac{1}{15}$$

e)

$$x + 3y - 4 = 0$$

$$3y = -x + 4$$

$$y = -\frac{1}{3}x + \frac{4}{3}$$

$$2x - 5y = 0$$

$$5y = 2x$$

$$y = \frac{2}{5}x$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{-\frac{1}{3} - \frac{2}{5}}{1 + (-\frac{1}{3})(\frac{2}{5})} \right|$$

$$= \frac{-11/15}{13/15}$$

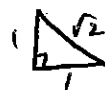
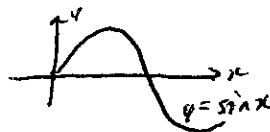
$$\tan \theta = 11/13$$

$$\theta = 40.236^\circ$$

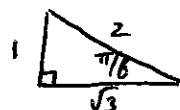
$$= 40^\circ \text{ to nearest degree}$$

Question Two

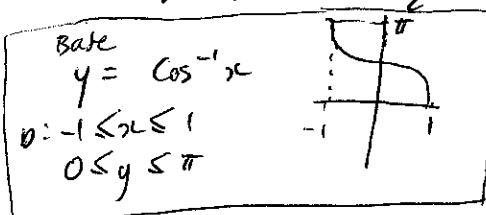
$$\begin{aligned}
 a) \int_0^{\pi/4} \cos^2 \frac{x}{2} dx &= \frac{1}{2} \int_0^{\pi/4} 1 + \cos x dx \\
 &= \frac{1}{2} \left[x + \sin x \right]_0^{\pi/4} \\
 &= \frac{1}{2} \left[\frac{\pi}{4} + \sin \frac{\pi}{4} - 0 \right] \\
 &= \frac{\pi}{8} + \frac{1}{2}\sqrt{2} \\
 &= \frac{\pi + 2\sqrt{2}}{8}
 \end{aligned}$$



$$\begin{aligned}
 b) \int_0^{1/5} \frac{2dx}{\sqrt{16-100x^2}} &= \int_0^{1/5} \frac{2dx}{2\sqrt{4-25x^2}} \\
 &= \int_0^{1/5} \frac{dx}{\sqrt{4/5-x^2}} \\
 &= \left[\sin^{-1} \frac{x}{2/5} \right]_0^{1/5} \\
 &= \sin^{-1} \frac{1}{2} - \sin^{-1} 0 \\
 &= \pi/6
 \end{aligned}$$

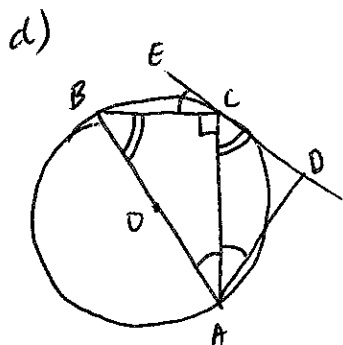
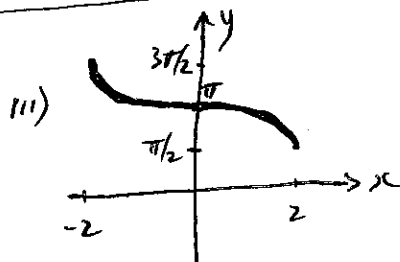


$$c) y = \frac{\pi}{2} + \cos^{-1} \frac{x}{2}$$



i) $\therefore \cos^{-1} \frac{x}{2}$ has domain $-1 \leq \frac{x}{2} \leq 1$
 $D: -2 \leq x \leq 2$

ii) $\frac{\pi}{2} + \cos^{-1} \frac{x}{2}$ range
 $y = \cos^{-1} \frac{x}{2}$ has range $0 \leq y \leq \pi$
 $\therefore \cos^{-1} \frac{x}{2} + \frac{\pi}{2}$ shifts up $\pi/2$.
 $R: \pi/2 \leq y \leq 3\pi/2$



$\angle BCA = 90^\circ$ angle in a semicircle.
 $\angle ACD = \angle ABC$ (angle between tangent and chord at point of contact equals angle in alternate segment.)
 $\angle BCE = \angle CAB$ point of contact equals angle in alternate segment.
 $\angle CAD = \angle BAC$ given AC bisects $\angle OAB$.

Question three

a) $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

LHS $\sin 3\theta = \sin(\theta + 2\theta)$
 $= \sin\theta \cos 2\theta + \cos\theta \sin 2\theta$
 $= \sin\theta(1 - 2\sin^2\theta) + \cos\theta(2\sin\theta \cos\theta)$
 $= \sin\theta - 2\sin^3\theta + 2\sin\theta \cos^2\theta$
 $= \sin\theta + 2\sin^3\theta + 2\sin\theta(1 - \sin^2\theta)$
 $= \sin\theta + 2\sin^3\theta + 2\sin\theta - 2\sin^3\theta$
 $= 3\sin\theta - 4\sin^3\theta$
 $= \text{RHS}$

d) $3\sin\theta - 4\sin^3\theta = -1$

$\sin 3\theta = -1$

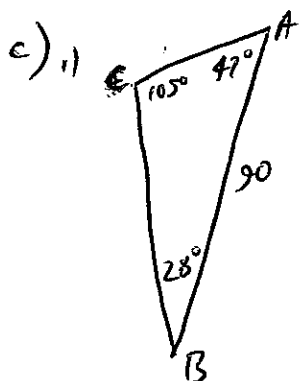
now if $0 \leq \theta \leq 2\pi$

then $0 \leq 3\theta \leq 6\pi$

$3\theta = \frac{3\pi}{2}, \frac{7\pi}{2}, \frac{11\pi}{2}, \frac{15\pi}{2}$
 $\theta = \pi/2, 7\pi/6, 11\pi/6$

b) $\frac{d}{dx} x \tan^{-1}x = \tan^{-1}x \cdot 1 + x \cdot \frac{1}{1+x^2}$
 $= \tan^{-1}x + \frac{x}{1+x^2}$

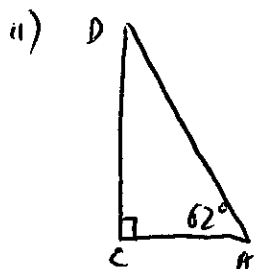
ii) $\tan^{-1}x = \frac{d}{dx}(x \tan^{-1}x) - \frac{x}{1+x^2}$
 $\therefore \int_0^1 \tan^{-1}x \, dx = \int_0^1 \frac{d}{dx}(x \tan^{-1}x) \, dx - \int_0^1 \frac{x}{1+x^2} \, dx$
 $= \left[x \tan^{-1}x \right]_0^1 - \left[\frac{1}{2} \ln(1+x^2) \right]_0^1$
 $= \left(1 \cdot \frac{\pi}{4} - 0 \right) - \left(\frac{1}{2} \ln 2 - \frac{1}{2} \ln 1 \right)$
 $= \frac{\pi}{4} - \frac{\ln 2}{2}$



$\frac{AC}{\sin 28} = \frac{90}{\sin 105}$

$AC = \frac{90 \times \sin 28}{\sin 105}$
 $= 43.74$

$= 44 \text{ m to nearest metre}$



$\tan 62 = \frac{DC}{AC}$

$DC = AC \tan 62$
 $= 82.26 \dots$

$= 82.3 \text{ to 3 sig. fig.}$

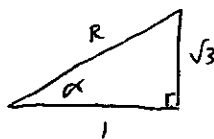
QUESTION 4

$$4a) i) \sqrt{3} \sin x - \cos x = R \sin(x - \alpha)$$

$$= R \sin x \cos \alpha - R \cos x \sin \alpha$$

$$\therefore R \sin \alpha = \sqrt{3} \quad \text{and} \quad R \cos \alpha = 1$$

$$\sin \alpha = \frac{\sqrt{3}}{R} \quad \cos \alpha = \frac{1}{R}$$



$$R = \sqrt{3+1}$$

$$= 2$$

$$\therefore \cos \alpha = \frac{1}{2}$$

$$\alpha = \frac{\pi}{6}$$

$$\therefore \sqrt{3} \sin x - \cos x = 2 \sin(x - \frac{\pi}{6})$$

$$ii) \sqrt{3} \sin x - \cos x = 1$$

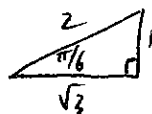
$$\therefore 2 \sin(x - \frac{\pi}{6}) = 1$$

$$\sin(x - \frac{\pi}{6}) = \frac{1}{2}$$

Now

$$0 \leq x \leq 2\pi$$

$$\therefore -\frac{\pi}{6} \leq x - \frac{\pi}{6} \leq \frac{11\pi}{6}$$



$$x - \frac{\pi}{6} = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}$$

$$x = \frac{\pi}{3}, \pi$$



$$4b) \left(2x - \frac{1}{x^3}\right)^{20} = \left(2x - x^{-3}\right)^{20}$$

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

$$= {}^{20}C_r 2x^{20-r} (-x^{-3})^r$$

$$= {}^{20}C_r 2^{20-r} x^{20-r} (-1)^r x^{-3r}$$

$$= {}^{20}C_r (-1)^r 2^{20-r} x^{20-4r}$$

for independent term. $20-4r=0$ ie $r=5$.

$$\therefore T_6 = {}^{20}C_5 (-1)^5 2^{15}$$

$$= - {}^{20}C_5 2^{15}$$

$$= - \frac{20!}{15!5!} 2^{15}$$

$$= -15504 \times 2^{15}$$

$$= -508035072$$

4c) i) $x^2 = 4ay$. when $x = 2at$. so $y - at^2 = -\frac{1}{t}(x - 2at)$
 $y = \frac{x^2}{4a}$ $y' = t$ $ty - at^3 = -x + 2at$
 $y' = \frac{x}{2a}$ $\therefore M_{\text{normal}} = -\frac{1}{t}$ $x + ty = 2at + at^3$

ii) to pass through the focus means it will pass through $(0, a)$
 ie $0 + at = 2at + at^3$
 $0 = at(1 + t^2)$ this has only one solution ie $t = 0$
 because $t^2 \neq -1$

d) let $f(x) = e^{-x^2} - 5x^2 - 0.99$
 $f'(x) = -2xe^{-x^2} - 10x$

Newton's method $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$

NOTE: $f'(0) = 0$ which will make this undefined.
 $f'(0) = 0$ cannot have gradient as it
 at $x=0$ ie. turning point at $x=0$.
 now $f(0) = e^0 - 0.99$
 $= 0.01$ so not a root either.

Question 5

a) step 1 prove true for $n=1$ i.e. $5^2 - 1 = 24$ which is divisible by 6.
 $\therefore$ true for $n=1$

step 2 assume true for $n=k$ i.e. $5^{2k} - 1 = 6M$ where M is an integer.

step 3 prove true for $n=k+1$ i.e. $5^{2k+2} - 1 = 5^2 5^{2k} - 1$

$$\text{from step 2 } 5^{2k} = 6M + 1.$$

$$= 25(6M + 1) - 1$$

$$= 25 \cdot 6M + 24.$$

both terms are divisible by 6.

$$= 6(25M + 4)$$

$\therefore$ true for $n=k+1$.

step 4 if true for $n=k$ then true for $n=k+1$.

since true for $n=1$. so true for $n=1+1$
 $= 2$ then $n=3$ and so on

for all integer values of n .

b) i) $V = \frac{4}{3} \pi r^3$

$$\frac{dV}{dr} = 4\pi r^2 \quad \text{now } \frac{dV}{dt} = 1000$$

$$\frac{dr}{dt} \times \frac{dV}{dr} = \frac{dV}{dt}$$

$$\frac{dr}{dt} \times 4\pi r^2 = 1000$$

$$\frac{dr}{dt} = \frac{1000}{4\pi r^2}$$

$$= \frac{250}{\pi r^2} \text{ cm/s}$$

ii) find $\frac{dA}{dt}$ when $r=10$

$$A = 4\pi r^2$$

$$\frac{dA}{dr} = 8\pi r$$

$$\therefore \frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}$$

$$= 8\pi r \times \frac{250}{\pi r^2}$$

$$= \frac{2000}{r}$$

$$\text{when } r=10 \quad \frac{dA}{dt} = 200 \text{ cm}^2/\text{s}$$

c) ${}^nC_r = {}^nC_{r+1}$

$$\frac{n!}{(n-r)! r!} = \frac{n!}{(n-r-1)! (r+1)!}$$

$$\frac{n!}{(n-r-1)! (n-r) r!} = \frac{n!}{(n-r-1)! r! (r+1)}$$

$$\frac{n! (r+1)}{(n-r-1)! (n-r) r! (r+1)} = \frac{n! (n-r)}{(n-r-1)! (n-r) r! (r+1)}$$

$\therefore r+1 = n-r$
 $n = 2r+1$ which is odd for all values of r as $2r$ is even positive integer.

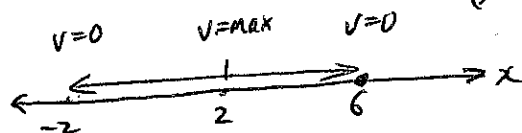
NOTE: $(n-r)! = (n-r-1)! (n-r)$

$(r+1)! = r! (r+1)$

Question 6 a)

i) $a = \frac{d \frac{1}{2} v^2}{dx}$ so $a = \frac{d(6+2x-\frac{x^2}{2})}{dx}$ for SHM $\ddot{x} = -\omega^2 x + k$
 $= 2 - x$
 $= -x^2 + 2 \therefore \text{SHM } \omega^2 = 1$

ii) at ends $v=0$ ie $v^2=0$ so $12+4x-x^2=0$ ends $x=-2$ and $x=6$
 $x^2-4x-12=0$ so centre = halfway
 $(x-6)(x+2)=0$ $x=2m$
 amplitude = centre to end
 $= 4m$



Period $T = \frac{2\pi}{\omega}$ and $\omega = 1 \therefore T = \frac{2\pi}{1}$
 $= 2\pi$ seconds.

iii) $x = a \sin(\omega t + \theta)$ is SHM about the centre $\therefore b = 2$
 $x = a \sin(\omega t + \theta) + b$ is SHM about pt. b now $a = 4, \omega = 1, b = 2$
 $x = 4 \sin(1t + \theta) + 2$

θ unknown but when $t=0$ $x=6$ (GIVEN)

$$6 = 4 \sin \theta + 2$$

$$4 = 4 \sin \theta$$

$$\sin \theta = 1 \therefore \theta = \pi/2$$

$$6b) \quad i) T = P + Ae^{kt} \quad \text{--- (1)}$$

$$\frac{dT}{dt} = kAe^{kt} \quad \text{now } Ae^{kt} = T - P \text{ from (1)}$$

$$= k(T - P)$$

$$ii) \quad P = 18 \quad t = 0 \quad T = 100$$

$$100 = 18 + Ae^0$$

$$A = 82$$

$$\therefore T = 18 + 82e^{kt}$$

$$\text{when } t = 3 \quad T = 87$$

$$87 = 18 + 82e^{3k}$$

$$e^{3k} = \frac{69}{82}$$

$$k = \frac{1}{3} \ln \frac{69}{82}$$

$$= -0.05753758$$

$$iii) \quad 70 = 18 + 82e^{-0.0575t}$$

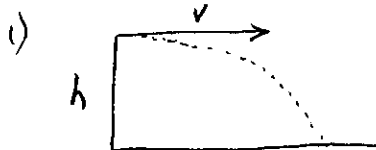
$$\ln \frac{52}{82} = -0.0575t$$

$$t = \frac{\ln \frac{52}{82}}{-0.0575}$$

$$= 7.916$$

$$= 8 \text{ min to nearest minute.}$$

QUESTION 7 a



$$\ddot{x} = 0$$

$$\dot{x} = C_1$$

= v since all horizontal

$$x = vt + C_2$$

$$\text{when } t=0 \quad x=0 \therefore C_2=0$$

$$\therefore x = vt$$

$$\ddot{y} = -g$$

$$\dot{y} = -gt + C_3$$

when $t=0$ vert. velocity = 0 so $C_3=0$

$$\therefore \dot{y} = -gt$$

$$y = -\frac{1}{2}gt^2 + C_4$$

$$\text{when } t=0 \quad y=h \therefore C_4=h$$

$$\therefore y = -\frac{1}{2}gt^2 + h$$

to be on the ground $y=0$

$$\therefore 0 = -\frac{1}{2}gt^2 + h$$

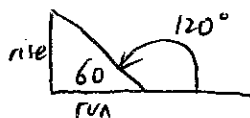
$$\frac{1}{2}gt^2 = h$$

$$t^2 = \frac{2h}{g}$$

$$t = \sqrt{\frac{2h}{g}} \quad \text{note: no negative time}$$

✓✓✓

ii) to strike ground at 60° ie 120°



hint: $\frac{dy}{dx} = \tan 120^\circ$
given

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

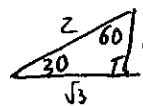
$$\text{now } \frac{dy}{dt} = -gt \quad \text{and } \frac{dx}{dt} = v \therefore \frac{dt}{dx} = \frac{1}{v}$$

$$= -gt \cdot \frac{1}{v}$$

strikes the ground at time $t = \sqrt{\frac{2h}{g}}$

$$\frac{dy}{dx} = -g \frac{\sqrt{\frac{2h}{g}}}{v}$$

$$\text{now } \tan 120^\circ = -\tan 60^\circ = -\sqrt{3}$$



$$\text{so } -\sqrt{3} = -g \frac{\sqrt{\frac{2h}{g}}}{v}$$

$$-\sqrt{3}v = -g \sqrt{\frac{2h}{g}}$$

$$3v^2 = g^2 \cdot \frac{2h}{g}$$

$$3v^2 = 2hg \quad \#$$

$$7b) i) y = \log_e(e^x \sin^2 x)$$

$$= \ln e^x + \ln \sin^2 x.$$

$$= x + 2 \ln \sin x$$

$$\frac{dy}{dx} = 1 + 2 \frac{\cos x}{\sin x}$$

$$= 1 + 2 \cot x. \quad \checkmark$$

$$ii) y = x + 2 \ln \sin x. \quad \text{when } x = \frac{\pi}{2}$$

$$\frac{dy}{dx} = 1 + 2 \cot x.$$

$$y = \frac{\pi}{2} + 2 \ln \sin \frac{\pi}{2}$$

$$= \frac{\pi}{2}$$

$$\frac{dy}{dx} = 1 + 2 \cot \frac{\pi}{2} \quad \text{so } M_{\text{normal}} = -1.$$

$$= 1$$

$\therefore$ eqⁿ of normal

$$y - \frac{\pi}{2} = -1 \left(x - \frac{\pi}{2} \right)$$

$$y - \frac{\pi}{2} = -x + \frac{\pi}{2}.$$

$$\boxed{x + y = \pi}$$

7c

$$x = 5 \tan \theta$$

$$x^2 = 25 \tan^2 \theta$$

$$\frac{dx}{d\theta} = 5 \sec^2 \theta$$

$$dx = 5 \sec^2 \theta d\theta.$$

$$\int \frac{x dx}{(25 + x^2)^{3/2}} = \int \frac{5 \tan \theta \cdot 5 \sec^2 \theta d\theta}{(25 + 25 \tan^2 \theta)^{3/2}}$$

$$= \int \frac{25 \tan \theta \sec^2 \theta d\theta}{25^{3/2} (1 + \tan^2 \theta)^{3/2}}$$

$$= \int \frac{25 \tan \theta \sec^2 \theta d\theta}{125 \sec^3 \theta} d\theta$$

$$= \frac{1}{5} \int \frac{\tan \theta}{\sec \theta} d\theta$$

$$= \frac{1}{5} \int \sin \theta d\theta.$$

$$= -\frac{1}{5} \cos \theta + C$$

$$= -\frac{1}{5} \cos \left[\cos^{-1} \frac{5}{\sqrt{x^2 + 25}} \right] + C.$$

$$= -\frac{1}{\sqrt{x^2 + 25}} + C.$$

$$(1 + \tan^2 \theta)^{3/2}$$

$$= (\sec^2 \theta)^{3/2}$$

$$= \sec^3 \theta.$$

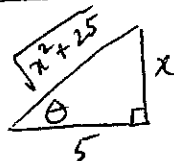
$$\frac{\tan \theta}{\sec \theta} = \frac{\sin \theta}{\frac{1}{\cos \theta}} \times \frac{\cos \theta}{1}$$

$$= \sin \theta.$$

now $x = 5 \tan \theta.$

$$\tan \theta = \frac{x}{5}.$$

$$\theta = \tan^{-1} \frac{x}{5}.$$

$$\tan \theta = \frac{x}{5}$$


$$\cos \theta = \frac{5}{\sqrt{x^2 + 25}} \quad \text{so } \theta = \cos^{-1} \frac{5}{\sqrt{x^2 + 25}}.$$